

PEKAN BARU SIMPANGTIGA, Indonesia

WMO: 961090

Lat: 0.461N

Lon: 101.445E

Elev: 102

StdP: 14.64

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	71.7	72.5	68.2	103.8	84.7	69.6	109.1	82.7	14.3	83.2	12.3	84.9	1.4	210	0.307

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	15.3	94.2	79.2	93.3	79.2	92.6	79.1	82.2	90.7	81.5	89.9	80.9	89.1	4.0	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.8	155.3	87.7	79.1	151.6	86.7	78.6	149.1	86.1	46.1	91.2	45.3	90.2	44.7	89.5	93.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.3	10.8	9.4	69.8	96.5	1.5	1.4	68.7	97.5	67.8	98.3	67.0	99.1	65.9	100.1		
(5)			67.5	85.4	2.5	2.9	65.7	87.5	64.2	89.2	62.8	90.9	61.0	93.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	82.4	81.2	81.8	82.6	83.3	83.6	83.4	82.8	82.6	82.1	82.4	82.0	81.4	(6)
(7)		DBStd	1.96	1.93	1.77	1.88	1.81	1.94	1.83	1.83	1.87	1.80	1.75	1.51	1.78	(7)
(8)		HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
(10)		CDD50	11842	966	891	1011	1000	1041	1002	1017	1011	964	1005	960	974	(10)
(11)		CDD65	6367	501	471	546	550	576	552	552	546	514	540	510	509	(11)
(12)		CDH74	64854	4597	4587	5525	5793	6375	6188	5958	5753	5157	5316	4876	4729	(12)
(13)		CDH80	26193	1581	1703	2257	2468	2758	2729	2553	2423	2073	2140	1852	1656	(13)
(14)	Wind	WSAvg	2.3	2.4	2.1	2.1	2.0	2.1	2.4	2.8	2.8	2.5	2.1	2.2	2.5	(14)
(15)	Precipitation	PrecAvg	102.5	7.4	7.8	9.9	10.3	8.9	6.4	5.0	6.9	8.5	9.4	12.5	10.6	(15)
(16)		PrecMax	128.4	19.6	20.9	18.1	16.5	15.1	14.5	10.2	16.4	15.0	17.4	24.3	21.8	(16)
(17)		PrecMin	56.4	1.7	1.2	3.1	4.1	1.8	1.2	1.3	1.3	0.6	2.6	4.4	4.1	(17)
(18)		PrecStd	14.8	4.0	4.8	4.1	3.3	3.6	3.1	2.5	3.9	3.7	3.8	4.3	3.7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	92.6	92.9	94.2	95.1	95.2	95.0	93.9	94.0	93.5	94.2	93.0	92.2	(19)
(20)			MCWB	77.2	78.6	79.0	80.0	80.2	79.1	78.7	78.4	78.9	78.2	79.0	77.9	(20)
(21)		2%	DB	90.8	91.4	92.8	93.3	93.5	93.3	92.8	92.8	92.1	92.3	91.4	90.8	(21)
(22)			MCWB	77.7	78.3	78.8	79.9	80.2	79.4	78.8	78.6	78.9	78.7	78.7	78.0	(22)
(23)		5%	DB	89.3	89.8	91.3	91.8	92.2	92.0	91.5	91.4	90.9	91.1	89.9	89.4	(23)
(24)			MCWB	77.7	78.0	78.9	79.7	80.1	79.3	78.7	78.6	78.8	78.8	78.7	78.0	(24)
(25)		10%	DB	87.7	88.2	89.7	90.4	90.8	90.9	90.0	89.9	89.4	89.5	88.6	87.7	(25)
(26)			MCWB	77.7	77.9	78.7	79.6	80.0	79.3	78.6	78.4	78.6	78.7	78.7	78.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	80.7	81.0	82.2	82.6	83.1	82.6	81.7	81.6	81.7	81.9	81.6	80.6	(27)
(28)			MCDWB	88.2	89.1	90.3	91.4	91.8	91.3	90.1	90.1	90.3	90.3	89.3	87.9	(28)
(29)		2%	WB	79.6	79.9	80.9	81.6	82.0	81.6	80.6	80.3	80.6	80.6	80.2	79.7	(29)
(30)			MCDWB	86.8	87.7	89.1	90.1	90.4	90.1	89.2	89.0	89.2	88.8	87.9	86.8	(30)
(31)		5%	WB	78.9	79.2	80.0	80.7	81.1	80.7	79.9	79.8	79.8	79.8	79.7	79.1	(31)
(32)			MCDWB	85.9	86.7	88.0	89.0	89.1	89.1	88.3	88.3	88.1	87.8	87.2	85.9	(32)
(33)		10%	WB	78.2	78.5	79.3	80.0	80.4	80.0	79.2	79.1	79.1	79.1	79.0	78.6	(33)
(34)			MCDWB	84.8	85.8	87.1	87.9	88.1	87.9	87.5	87.5	87.0	86.9	86.5	85.2	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	13.4	14.4	15.4	15.6	15.3	15.7	15.8	15.7	15.2	15.0	14.5	13.6	(35)
(36)			MCDBR	16.0	16.3	17.5	17.3	17.1	17.3	17.2	17.5	17.3	17.0	16.2	15.8	(36)
(37)			MCWBR	5.5	5.7	6.3	6.2	6.2	6.2	6.0	6.1	6.5	6.0	5.9	5.5	(37)
(38)		5% WB	MCDWR	13.8	14.8	16.0	15.9	15.8	16.0	16.1	16.1	16.0	15.4	14.8	14.1	(38)
(39)			MCWBR	5.5	5.9	6.6	6.6	6.7	6.7	6.5	6.7	7.0	6.4	6.1	5.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.477	0.592	0.556	0.473	0.468	0.495	0.488	0.554	0.593	0.619	0.476	0.443	(40)	
(41)		taud	2.229	1.887	1.999	2.250	2.271	2.182	2.206	2.003	1.900	1.820	2.244	2.351	(41)	
(42)		Ebn at Noon	272	243	251	266	260	249	252	242	238	233	270	280	(42)	
(43)		Edh at Noon	47	67	59	45	43	45	45	56	64	70	46	41	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1381	1434	1550	1590	1557	1526	1527	1513	1484	1501	1484	1383	(44)	
(45)		RadStd	93	71	99	43	64	61	62	87	135	190	52	110	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	+0.58	+0.61	N/A	+0.58	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page